Endocrine & Exocrine Dysfunction After Pancreaticoduodenectomy

Combined Results Tables — Generated: May 06, 2026

Table1 Patient Preoperative Characteristics

Table 1. Patient and Preoperative Characteristics of 742 Patients Undergoing Pancreaticoduodenectomy

Characteristic	No.	Median (IQR) or No. (%)
Demographics		
Age at surgery, years	742	66.9 (58.2–74.4)
Female sex	742	397 (53.5%)
Race/ethnicity	742	
Asian		9 (1.2%)
Black		50 (6.7%)
Hispanic		2 (0.3%)
Other		20 (2.7%)
Pacific		1 (0.1%)
White		660 (88.9%)
Body mass index, kg/m²	742	25.8 (23.1–29.4)
Pre-operative Comorbidities		
Hypertension	742	387 (52.2%)
Hyperlipidemia	742	326 (43.9%)
Hepatic fibrosis or cirrhosis	742	13 (1.8%)
History of significant alcohol use	742	205 (27.6%)
Tobacco use	742	
current		93 (12.5%)
never		364 (49.1%)
past		285 (38.4%)
History of pancreatitis	742	
acute		95 (12.8%)
chronic		83 (11.2%)
none		564 (76.0%)
Proton pump inhibitor use	742	258 (34.8%)
Chronic steroid use	742	40 (5.4%)
Prior gastrointestinal malignancy	742	40 (5.4%)
Other prior malignancy	742	179 (24.1%)
Family history of diabetes mellitus	742	212 (28.6%)
Pre-operative Glycemic Status		
HbA1c, %	222	5.5 (5.3–5.9)
HbA1c category	222	

Abbreviations: Data are presented as median (interquartile range) for continuous variables and as number (percentage) for categorical variables. BMI = body mass index; EUS = endoscopic ultrasound; FNA = fine needle aspiration; EBL = estimated blood loss; CBD = common bile duct; TPN = total parenteral nutrition; LVI = lymphovascular invasion; PNI = perineural invasion; ISGPS = International Study Group on Pancreatic Surgery; POPF = postoperative pancreatic fistula; DGE = delayed gastric emptying; PPH = post-pancreatectomy hemorrhage; PERT = pancreatic enzyme replacement therapy; DM = diabetes mellitus; AJCC = American Joint Committee on Cancer; IQR = interquartile range.

Characteristic	No.	Median (IQR) or No. (%)
Diabetic range (≥6.5%)		11 (5.0%)
Normal (<5.7%)		129 (58.1%)
Prediabetes (5.7–6.4%)		82 (36.9%)
Pre-operative Diagnostic Workup		
Endoscopic ultrasound (EUS)	742	589 (79.4%)
Fine needle aspiration (FNA)	742	465 (62.7%)
Computed tomography (CT)	742	523 (70.5%)
Magnetic resonance imaging (MRI)	742	663 (89.4%)
Positron emission tomography (PET)	742	71 (9.6%)
Index lesion size, cm	664	2.4 (1.6–3.4)
Indication for Pancreaticoduodenectomy		
Primary indication	741	
Pancreatitis		55 (7.4%)
Pancreatitis (DP)		26 (3.5%)
Tumor		472 (63.7%)
Tumor (DP)		188 (25.4%)

Abbreviations: Data are presented as median (interquartile range) for continuous variables and as number (percentage) for categorical variables. BMI = body mass index; EUS = endoscopic ultrasound; FNA = fine needle aspiration; EBL = estimated blood loss; CBD = common bile duct; TPN = total parenteral nutrition; LVI = lymphovascular invasion; PNI = perineural invasion; ISGPS = International Study Group on Pancreatic Surgery; POPF = postoperative pancreatic fistula; DGE = delayed gastric emptying; PPH = post-pancreatectomy hemorrhage; PERT = pancreatic enzyme replacement therapy; DM = diabetes mellitus; AJCC = American Joint Committee on Cancer; IQR = interquartile range.

Table2 Operative Pathologic Outcomes

Table 2. Operative, Pathologic, and Postoperative Outcomes of 742 Patients Undergoing Pancreaticoduodenectomy

Characteristic	No.	Median (IQR) or No. (%)
Operative Characteristics		
Resection type	742	
Central		4 (0.5%)
Distal		249 (33.6%)
PPPD		415 (55.9%)
sPD		74 (10.0%)
Operative approach	735	
laparoscopic		353 (48.0%)
open		360 (49.0%)
robotic		22 (3.0%)
ASA physical status classification	739	
I		3 (0.4%)
II		189 (25.6%)
III		524 (70.9%)
IV		23 (3.1%)
Operative time, minutes	741	257.0 (174.0–359.0)
Estimated blood loss, mL	733	150.0 (50.0–275.0)
Venous resection and reconstruction	742	63 (8.5%)
Concomitant splenectomy	742	243 (32.7%)
Concomitant organ resection	742	152 (20.5%)
Pancreatic gland texture	0	
Main pancreatic duct diameter, mm	452	3.0 (2.0–4.0)
Common bile duct diameter, mm	428	10.0 (8.0–15.0)
Intraoperative drain placement	742	95 (12.8%)
Enteral feeding tube placement	742	24 (3.2%)
Total parenteral nutrition	742	65 (8.8%)
Postoperative Course		
ICU length of stay, days	740	0.0 (0.0–0.0)
Hospital length of stay, days	740	5.0 (4.0–7.0)
Unplanned readmission within 90 days	558	131 (23.5%)
Reoperation	740	29 (3.9%)
Postoperative Complications		

Abbreviations: Data are presented as median (interquartile range) for continuous variables and as number (percentage) for categorical variables. BMI = body mass index; EUS = endoscopic ultrasound; FNA = fine needle aspiration; EBL = estimated blood loss; CBD = common bile duct; TPN = total parenteral nutrition; LVI = lymphovascular invasion; PNI = perineural invasion; ISGPS = International Study Group on Pancreatic Surgery; POPF = postoperative pancreatic fistula; DGE = delayed gastric emptying; PPH = post-pancreatectomy hemorrhage; PERT = pancreatic enzyme replacement therapy; DM = diabetes mellitus; AJCC = American Joint Committee on Cancer; IQR = interquartile range.

Characteristic	No.	Median (IQR) or No. (%)
Clinically relevant POPF	741	89 (12.0%)
POPF grade (ISGPS)	89	
Biochemical leak		26 (29.2%)
Grade B		55 (61.8%)
Grade C		8 (9.0%)
Delayed gastric emptying	742	78 (10.5%)
DGE grade (ISGPS)	76	
A		22 (28.9%)
B		26 (34.2%)
C		28 (36.8%)
Post-pancreatectomy hemorrhage	742	41 (5.5%)
PPH grade (ISGPS)	42	
A		10 (23.8%)
B		16 (38.1%)
C		16 (38.1%)
Surgical site infection	739	44 (6.0%)
Intra-abdominal abscess	742	72 (9.7%)
Clavien-Dindo severity grade	317	
Death		18 (5.7%)
I		66 (20.8%)
II		106 (33.4%)
IIIa		78 (24.6%)
IIIb		16 (5.0%)
IVa		20 (6.3%)
IVb		13 (4.1%)
Pathologic Findings		
Primary pathologic diagnosis	737	
Adenocarcinoma (Panc/CBD/Ampulla/Duodenum)		467 (63.4%)
Cystic disease (IPMN/MCN/SCN)		153 (20.8%)
Miscellaneous periampullary		36 (4.9%)
Pancreatitis		81 (11.0%)
Tumor size, cm	688	2.5 (1.5–3.5)
Surgical margin status	711	
R0		666 (93.7%)
R1		40 (5.6%)

Abbreviations: Data are presented as median (interquartile range) for continuous variables and as number (percentage) for categorical variables. BMI = body mass index; EUS = endoscopic ultrasound; FNA = fine needle aspiration; EBL = estimated blood loss; CBD = common bile duct; TPN = total parenteral nutrition; LVI = lymphovascular invasion; PNI = perineural invasion; ISGPS = International Study Group on Pancreatic Surgery; POPF = postoperative pancreatic fistula; DGE = delayed gastric emptying; PPH = post-pancreatectomy hemorrhage; PERT = pancreatic enzyme replacement therapy; DM = diabetes mellitus; AJCC = American Joint Committee on Cancer; IQR = interquartile range.

Characteristic	No.	Median (IQR) or No. (%)
R2		5 (0.7%)
Lymphovascular invasion	723	164 (22.7%)
Perineural invasion	723	287 (39.7%)
Positive lymph nodes, n	733	0.0 (0.0–2.0)
Total lymph nodes examined, n	730	20.5 (14.0–27.0)
AJCC/NCCN tumor stage	484	
Stage 0		10 (2.1%)
Stage IA		61 (12.6%)
Stage IB		79 (16.3%)
Stage IIA		55 (11.4%)
Stage IIB		144 (29.8%)
Stage III		135 (27.9%)
Endocrine Outcomes		
New-onset diabetes mellitus	741	154 (20.8%)
Time to diabetes diagnosis, days	154	738.0 (179.5–1600.8)
Insulin therapy initiated	742	95 (12.8%)
Exocrine Outcomes		
Pancreatic exocrine insufficiency	742	378 (50.9%)
Time to exocrine insufficiency, days	379	45.0 (-28.0–240.0)
Pancreatic enzyme replacement (PERT)	742	386 (52.0%)
Cholestyramine prescribed	742	62 (8.4%)
Abdominal pain	742	495 (66.7%)
diarrhea	742	379 (51.1%)
Follow-up		
Follow-up duration, months	742	34.7 (14.5–75.5)
Vital status at last follow-up	739	
Alive		441 (59.7%)
Dead		298 (40.3%)
Disease recurrence	504	184 (36.5%)

Abbreviations: Data are presented as median (interquartile range) for continuous variables and as number (percentage) for categorical variables. BMI = body mass index; EUS = endoscopic ultrasound; FNA = fine needle aspiration; EBL = estimated blood loss; CBD = common bile duct; TPN = total parenteral nutrition; LVI = lymphovascular invasion; PNI = perineural invasion; ISGPS = International Study Group on Pancreatic Surgery; POPF = postoperative pancreatic fistula; DGE = delayed gastric emptying; PPH = post-pancreatectomy hemorrhage; PERT = pancreatic enzyme replacement therapy; DM = diabetes mellitus; AJCC = American Joint Committee on Cancer; IQR = interquartile range.

Table3 Endocrine Exocrine Dysfunction

Table 3. Incidence and Timing of Endocrine and Exocrine Dysfunction Following Pancreatectomy (N = 742)

		Endocrine Outcomes			Exocrine Outcomes		
Group	**N**	**New-onset DM No. (%)**	**Time to DM Months, Median (IQR)**	**Insulin Initiated No. (%)**	**Exocrine Insufficiency No. (%)**	**Time to Exo Insuff Months, Median (IQR)**	**PERT Initiated No. (%)**
Overall	742	154 (20.8%)	24.4 (5.9–53.7)	95 (12.8%)	378 (50.9%)	5.0 (1.5–14.0)	386 (52.0%)
Stratified by Resection Type							
PPPD	415	68 (16.4%)	25.2 (6.7–50.5)	41 (9.9%)	241 (58.1%)	5.3 (1.6–13.9)	247 (59.5%)
Standard PD	74	15 (20.3%)	8.7 (2.8–15.3)	10 (13.5%)	49 (66.2%)	4.0 (1.2–7.1)	49 (66.2%)
Distal	249	69 (27.7%)	28.6 (6.5–56.6)	42 (16.9%)	87 (34.9%)	3.4 (1.2–20.9)	89 (35.7%)

Abbreviations and Notes: Data are presented as number (percentage) for incidence outcomes and median (interquartile range) in months for time-to-event outcomes. Time-to-event calculated from date of surgery to first documented diagnosis. Patients with pre-operative exocrine insufficiency were retained and flagged as a covariate (n = 123). Resection groups with fewer than 10 patients are not shown separately. DM = diabetes mellitus; PERT = pancreatic enzyme replacement therapy; IQR = interquartile range; PPPD = pylorus-preserving pancreaticoduodenectomy.

Table4A Univariable DM

Table 4A. Univariable Logistic Regression for Predictors of New-Onset Diabetes Mellitus Following Pancreatectomy (Events: 154/741 [20.8%])

Variable	N	OR	95% CI	P-Value
Demographics				
Age at surgery (per year)	741	1.00	0.98–1.01	0.570
Female sex	741	1.31	0.92–1.88	0.133
BMI (per unit, continuous)	741	1.00	NA–1.00	0.879
BMI category	741			
Normal (18.5-24.9) (reference)		1.00	—	—
Underweight (<18.5)		0.43	0.02–2.21	0.417
Overweight (25-29.9)		1.68	1.09–2.61	0.020 *
Obese (>=30)		2.77	1.75–4.43	<0.001 *
Pre-operative Glycemic Status				
Pre-op HbA1c (continuous)	222	2.96	1.70–5.48	<0.001 *
HbA1c category	211			
Normal (<5.7%) (reference)		1.00	—	—
Prediabetes (5.7-6.4)		3.11	1.65–5.96	<0.001 *
Pre-operative Comorbidities				
Hypertension	741	1.10	0.77–1.57	0.615
Hyperlipidemia	741	1.27	0.89–1.81	0.187
Family history of DM	741	1.41	1.05–1.89	0.020 *
History of pancreatitis	741	1.37	0.91–2.03	0.126
Significant alcohol use	741	1.45	0.98–2.11	0.058 *
Tobacco use (ever)	741	1.47	1.03–2.11	0.035 *
Pre-op exocrine insufficiency	741	1.36	0.85–2.11	0.187
Surgical Characteristics				
Distal pancreatectomy	741	1.84	1.28–2.64	0.001 *
Soft gland texture (score 1-5)	551	0.76	0.49–1.18	0.216
Main pancreatic duct diameter (per mm)	451	0.97	0.83–1.12	0.718
Vein resection	741	0.89	0.44–1.66	0.723
Estimated blood loss (per 100 mL)	732	1.00	0.97–1.03	0.849
Operative time (per 10 min)	740	1.00	0.99–1.02	0.644

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 4A: Odds ratios estimated by standard binary logistic regression; * denotes p < 0.10. Table 4B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes p < 0.05. Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 206). Patients with pre-operative HbA1c >=6.5% were excluded from the analytic cohort (n = 11) as this value meets the American Diabetes Association diagnostic threshold for diabetes mellitus irrespective of formal clinical diagnosis. Resection type (distal, total pancreatectomy) was excluded from both multivariable models due to complete separation; univariable estimates are reported in Table 4A. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%). HbA1c category in Model B includes Normal (<5.7%) and Prediabetes (5.7-6.4%) only; the dysglycemic range category (>=6.5%) was removed following cohort exclusion.

Variable	N	OR	95% CI	P-Value
Postoperative Complications				
Any POPF	740	1.31	0.77–2.18	0.304
Clinically relevant POPF (Grade B/C)	740	1.36	0.73–2.43	0.313
Delayed gastric emptying	741	0.91	0.49–1.61	0.766
Pathologic Characteristics				
Adenocarcinoma diagnosis	736	0.61	0.42–0.88	0.007 *
Tumor size (per cm)	687	0.98	0.90–1.07	0.686
R0 resection margin	710	0.78	0.40–1.65	0.490
Perineural invasion	722	0.70	0.48–1.02	0.067 *
Lymphovascular invasion	722	0.69	0.43–1.08	0.112
Oncologic Treatment				
Neoadjuvant therapy	741	0.64	0.39–1.00	0.059 *
Adjuvant chemotherapy	188	1.01	0.35–3.66	0.990

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 4A: Odds ratios estimated by standard binary logistic regression; * denotes p < 0.10. Table 4B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes p < 0.05. Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 206). Patients with pre-operative HbA1c >=6.5% were excluded from the analytic cohort (n = 11) as this value meets the American Diabetes Association diagnostic threshold for diabetes mellitus irrespective of formal clinical diagnosis. Resection type (distal, total pancreatectomy) was excluded from both multivariable models due to complete separation; univariable estimates are reported in Table 4A. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%). HbA1c category in Model B includes Normal (<5.7%) and Prediabetes (5.7-6.4%) only; the dysglycemic range category (>=6.5%) was removed following cohort exclusion.

Table4B Multivariable DM

Table 4B. Multivariable Logistic Regression for Predictors of New-Onset Diabetes Mellitus Following Pancreatectomy (Events: 154/741 [20.8%]). Model A: full cohort (N = 735); Model B: HbA1c subcohort (N = 206).

Variable	N	OR	95% CI	P-Value
Model A — Full cohort (no HbA1c)	**735**			
Age at surgery (per year)		1.01	0.99–1.02	0.401
BMI: Underweight vs Normal		0.57	0.06–2.58	0.510
BMI: Overweight vs Normal		1.90	1.22–3.00	0.005 *
BMI: Obese vs Normal		2.92	1.81–4.73	<0.001 *
Family history of DM		1.28	0.94–1.73	0.122
Clinically relevant POPF		1.21	0.64–2.21	0.542
Neoadjuvant therapy		0.87	0.50–1.47	0.595
Adenocarcinoma diagnosis		0.66	0.42–1.02	0.061
History of pancreatitis		1.20	0.75–1.89	0.451
Pre-op exocrine insufficiency		1.49	0.89–2.48	0.131
Tobacco use (ever)		1.49	1.03–2.17	0.036 *
Model B — HbA1c subcohort	**206**			
Age at surgery (per year)		1.00	0.97–1.03	0.985
BMI: Underweight vs Normal		0.84	0.01–9.23	0.909
BMI: Overweight vs Normal		1.61	0.68–3.84	0.278
BMI: Obese vs Normal		2.48	1.11–5.73	0.027 *
HbA1c: Prediabetes vs Normal (<5.7%)		3.07	1.58–6.11	<0.001 *
Family history of DM		0.86	0.49–1.48	0.597
Clinically relevant POPF		0.75	0.19–2.57	0.661
Neoadjuvant therapy		1.83	0.66–5.03	0.242
Adenocarcinoma diagnosis		0.28	0.11–0.66	0.003 *
History of pancreatitis		1.01	0.46–2.14	0.987
Pre-op exocrine insufficiency		1.60	0.68–3.68	0.273

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 4A: Odds ratios estimated by standard binary logistic regression; * denotes p < 0.10. Table 4B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes p < 0.05. Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 206). Patients with pre-operative HbA1c >=6.5% were excluded from the analytic cohort (n = 11) as this value meets the American Diabetes Association diagnostic threshold for diabetes mellitus irrespective of formal clinical diagnosis. Resection type (distal, total pancreatectomy) was excluded from both multivariable models due to complete separation; univariable estimates are reported in Table 4A. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%). HbA1c category in Model B includes Normal (<5.7%) and Prediabetes (5.7-6.4%) only; the dysglycemic range category (>=6.5%) was removed following cohort exclusion.

Table5A Univariable Exocrine

Table 5A. Univariable Logistic Regression for Predictors of Post-Operative Exocrine Insufficiency Following Pancreatectomy (Events: 261/619 [42.2%])

Variable	N	OR	95% CI	P-Value
Demographics				
Age at surgery (per year)	619	1.00	0.99–1.01	0.992
Female sex	619	1.16	0.84–1.60	0.357
BMI (per unit, continuous)	619	0.99	0.96–1.00	0.568
BMI category	619			
Normal (18.5-24.9) (reference)		1.00	—	—
Underweight (<18.5)		0.38	0.06–1.59	0.228
Overweight (25-29.9)		0.98	0.68–1.41	0.904
Obese (>=30)		0.91	0.59–1.38	0.651
Pre-operative Glycemic Status				
Pre-op HbA1c (continuous)	176	0.94	0.56–1.59	0.824
HbA1c category	168			
Normal (<5.7%) (reference)		1.00	—	—
Prediabetes (5.7-6.4)		0.85	0.45–1.59	0.607
Pre-operative Comorbidities				
Hypertension	619	0.85	0.62–1.17	0.316
Hyperlipidemia	619	0.96	0.70–1.32	0.808
Family history of DM	619	1.07	0.81–1.42	0.627
History of pancreatitis	619	0.99	0.65–1.48	0.945
Significant alcohol use	619	1.12	0.78–1.61	0.528
Tobacco use (ever)	619	0.99	0.72–1.36	0.928
Surgical Characteristics				
Distal pancreatectomy	619	0.34	0.23–0.48	<0.001 *
Soft gland texture (score 1-5)	454	0.47	0.32–0.71	<0.001 *
Main pancreatic duct diameter (per mm)	368	1.08	0.95–1.22	0.244
Vein resection	619	1.53	0.88–2.70	0.134
Estimated blood loss (per 100 mL)	611	1.00	0.97–1.03	0.963
Operative time (per 10 min)	618	1.03	1.01–1.04	<0.001 *
Postoperative Complications				
Any POPF	618	1.20	0.74–1.93	0.454

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 5A: Odds ratios estimated by standard binary logistic regression; * denotes p < 0.10. Table 5B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes p < 0.05. Patients with pre-operative exocrine insufficiency were excluded from the outcome analysis (n = 123). Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 164). New-onset diabetes mellitus was included as a time-concurrent covariate reflecting shared pathophysiologic mechanisms of pancreatic parenchymal loss. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%).

Variable	N	OR	95% CI	P-Value
Clinically relevant POPF (Grade B/C)	618	1.26	0.72–2.17	0.411
Delayed gastric emptying	619	1.23	0.72–2.08	0.439
New-onset diabetes mellitus	618	2.02	1.35–3.02	<0.001 *
Pathologic Characteristics				
Adenocarcinoma diagnosis	615	2.62	1.85–3.75	<0.001 *
Tumor size (per cm)	583	0.90	0.83–0.97	0.011 *
R0 resection margin	593	0.53	0.27–1.02	0.059 *
Perineural invasion	602	1.69	1.22–2.36	0.002 *
Lymphovascular invasion	602	1.07	0.73–1.56	0.721
Oncologic Treatment				
Neoadjuvant therapy	619	2.45	1.62–3.74	<0.001 *
Adjuvant chemotherapy	136	1.16	0.46–2.87	0.745

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 5A: Odds ratios estimated by standard binary logistic regression; * denotes $p < 0.10$. Table 5B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes $p < 0.05$. Patients with pre-operative exocrine insufficiency were excluded from the outcome analysis (n = 123). Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 164). New-onset diabetes mellitus was included as a time-concurrent covariate reflecting shared pathophysiologic mechanisms of pancreatic parenchymal loss. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%).

Table5B Multivariable Exocrine

Table 5B. Multivariable Logistic Regression for Predictors of Post-Operative Exocrine Insufficiency Following Pancreatectomy (Events: 261/619 [42.2%]). Model A: full cohort (N = 613); Model B: HbA1c subcohort (N = 164).

Variable	N	OR	95% CI	P-Value
Model A — Full cohort	613			
Age at surgery (per year)		0.99	0.97–1.00	0.107
BMI: Underweight vs Normal		0.29	0.05–1.20	0.089
BMI: Overweight vs Normal		0.80	0.53–1.18	0.257
BMI: Obese vs Normal		0.78	0.48–1.23	0.284
Family history of DM		1.21	0.89–1.65	0.228
Clinically relevant POPF		1.28	0.71–2.32	0.405
Neoadjuvant therapy		1.80	1.13–2.89	0.013 *
Adenocarcinoma diagnosis		2.06	1.32–3.25	0.001 *
History of pancreatitis		1.03	0.65–1.62	0.890
Tobacco use (ever)		0.91	0.64–1.29	0.597
Distal pancreatectomy		0.42	0.28–0.63	<0.001 *
New-onset diabetes mellitus		2.67	1.72–4.20	<0.001 *
Model B — HbA1c subcohort	164			
Age at surgery (per year)		0.98	0.95–1.01	0.130
BMI: Underweight vs Normal		0.23	0.02–2.17	0.196
BMI: Overweight vs Normal		0.84	0.36–1.92	0.675
BMI: Obese vs Normal		0.33	0.14–0.77	0.010 *
HbA1c: Prediabetes vs Normal (<5.7%)		0.83	0.41–1.68	0.607
Family history of DM		0.99	0.58–1.68	0.962
Clinically relevant POPF		2.49	0.67–9.55	0.170
Neoadjuvant therapy		3.74	1.35–11.88	0.010 *
Adenocarcinoma diagnosis		3.24	1.45–7.67	0.004 *
History of pancreatitis		1.48	0.66–3.33	0.339
New-onset diabetes mellitus		1.90	0.82–4.57	0.136

Abbreviations and Notes: OR = odds ratio; CI = confidence interval. Table 5A: Odds ratios estimated by standard binary logistic regression; * denotes p < 0.10. Table 5B: Odds ratios estimated by Firth penalized likelihood regression to address sparse data; profile likelihood confidence intervals reported; * denotes p < 0.05. Patients with pre-operative exocrine insufficiency were excluded from the outcome analysis (n = 123). Model A was fitted in the full analytic cohort, excluding predictors with >25% missing data (pre-operative HbA1c, pancreatic duct diameter, gland texture). Model B was restricted to patients with available pre-operative HbA1c (n = 164). New-onset diabetes mellitus was included as a time-concurrent covariate reflecting shared pathophysiologic mechanisms of pancreatic parenchymal loss. Reference categories: BMI, Normal (18.5-24.9 kg/m²); HbA1c, Normal (<5.7%).

Table6 HbA1c Trajectory

Table 6. HbA1c Trajectory Following Pancreatectomy Stratified by New-Onset Diabetes Status (New-Onset DM: n = 154; No DM: n = 587)

	New-Onset DM		No DM		
Time Point	**N**	**New-Onset DM Median HbA1c, % (IQR)**	**N**	**No DM Median HbA1c, % (IQR)**	**P-Value[1]**
Pre-operative	60	5.8 (5.4–6.2)	162	5.5 (5.2–5.9)	<0.001
Year 1	61	6.3 (6.0–6.9)	98	5.5 (5.2–5.9)	<0.001
Year 2	41	6.8 (6.0–8.0)	46	5.6 (5.4–6.0)	<0.001
Year 3	45	6.7 (6.1–7.6)	40	5.6 (5.5–5.9)	<0.001
Year 4	45	6.9 (6.2–7.6)	35	5.7 (5.4–5.9)	<0.001
Year 5	41	6.7 (6.2–7.6)	37	5.6 (5.2–5.9)	<0.001
Year 6	38	6.7 (6.1–8.0)	24	5.8 (5.4–5.9)	<0.001
Year 7	30	7.0 (6.2–7.6)	19	5.7 (5.6–5.9)	<0.001
Year 8	25	7.0 (6.1–9.0)	20	5.6 (5.2–5.8)	<0.001
Year 9	22	7.2 (6.4–8.1)	17	5.9 (5.6–6.1)	<0.001
Year 10	19	7.3 (6.0–8.6)	14	5.8 (5.7–6.2)	0.005

Abbreviations:

HbA1c values reported as median (interquartile range) in percent (%). N denotes the number of patients with available HbA1c measurement at each time point. High rates of missingness at later time points reflect loss to follow-up and mortality, particularly in the adenocarcinoma subgroup. Pre-operative HbA1c excludes patients with values >=6.5% (n = 11) who were removed from the analytic cohort. IQR = interquartile range; DM = diabetes mellitus.

[1] P-value from two-sided Wilcoxon rank-sum test comparing HbA1c values between groups at each time point. Time points with fewer than 3 observations in either group are suppressed (—).

Table7 LMM HbA1c Trajectory

Table 7. Linear Mixed Model Fixed Effects for Post-Operative HbA1c Trajectory Following Pancreatectomy

Variable	β	SE	95% CI	P-Value
Model 1 — Unadjusted (N obs = 700; N patients = 306)				
Group and Time Effects				
Intercept	7.321	0.768	5.816–8.826	<0.001 *
Time (per year after Y1)	-0.289	0.154	-0.591–0.013	0.063 †
No DM — Prediabetes vs Normal HbA1c	-1.411	1.380	-4.116–1.294	0.308
No DM — HbA1c unknown vs Normal HbA1c	-1.768	0.913	-3.557–0.020	0.054 †
New-Onset DM vs Normal HbA1c	-0.660	0.921	-2.466–1.146	0.474
Time × Group Interactions				
Time × No DM — Prediabetes	0.274	0.269	-0.253–0.800	0.311
Time × No DM — HbA1c unknown	0.300	0.182	-0.056–0.656	0.102
Time × New-Onset DM	0.408	0.181	0.053–0.763	0.026 *
Model 2A — Adjusted, full cohort (N obs = 696; N patients = 305)				
Group and Time Effects				
Intercept	6.072	1.582	2.970–9.173	<0.001 *
Time (per year after Y1)	-0.001	0.060	-0.118–0.116	0.988
No DM — Prediabetes vs Normal HbA1c	-1.774	1.302	-4.327–0.778	0.174
No DM — HbA1c unknown vs Normal HbA1c	-1.609	0.867	-3.310–0.091	0.065 †
New-Onset DM vs Normal HbA1c	-0.740	0.891	-2.486–1.005	0.406
Time × Group Interactions				
Time × No DM — Prediabetes	0.022	0.138	-0.248–0.291	0.876
Time × No DM — HbA1c unknown	0.022	0.078	-0.131–0.175	0.777
Time × New-Onset DM	0.132	0.066	0.002–0.262	0.048 *
Baseline Covariates				
Pre-op HbA1c (per 1%)	—	—	—	—
Age at surgery (per year)	0.010	0.023	-0.034–0.055	0.648
BMI: Underweight vs Normal	0.564	3.527	-6.350–7.477	0.873

Abbreviations and Notes: β = unstandardized regression coefficient (HbA1c %, change per unit). SE = standard error. CI = 95% confidence interval (Wald). Models fitted by restricted maximum likelihood (REML) using lme4. P-values from Satterthwaite degrees of freedom approximation (lmerTest). * $p < 0.05$; † $p < 0.10$. Time is centered at Year 1 (time_c = 0 corresponds to Year 1 post-op). The β for Time represents the annual HbA1c change in the reference group (No DM — Normal HbA1c). Time × Group interaction terms represent the additional annual change in each group relative to the reference. Model 1: unadjusted, random intercept + slope. Model 2A: adjusted, random intercept, full cohort (pre-op HbA1c excluded due to 56% missingness). Model 2B: adjusted, random intercept, restricted to patients with available pre-op HbA1c. The No DM — Prediabetes group is included in model estimation but suppressed from the trajectory figure due to unreliable estimated marginal means arising from limited post-operative HbA1c observations (n ≈ 50). Reference categories: DM/glycemic group, No DM — Normal HbA1c; BMI, Normal (18.5–24.9 kg/m²). Pre-operative HbA1c ≥6.5% excluded (n = 11, ADA DM criteria).

Variable	β	SE	95% CI	P-Value
BMI: Overweight vs Normal	0.317	0.654	-0.965–1.600	0.628
BMI: Obese vs Normal	1.245	0.755	-0.235–2.724	0.100
Distal pancreatectomy	0.135	0.633	-1.106–1.375	0.832
Neoadjuvant therapy	-0.442	0.808	-2.025–1.141	0.585
Adenocarcinoma diagnosis	0.488	0.718	-0.919–1.895	0.497
Tobacco use (ever)	-0.288	0.575	-1.414–0.839	0.617
Model 2B — Adjusted, HbA1c subcohort (N obs = 303; N patients = 109)				
Group and Time Effects				
Intercept	-3.518	12.696	-28.402–21.365	0.782
Time (per year after Y1)	0.008	0.064	-0.117–0.133	0.899
No DM — Prediabetes vs Normal HbA1c	-3.028	2.690	-8.301–2.245	0.263
No DM — HbA1c unknown vs Normal HbA1c	—	—	—	—
New-Onset DM vs Normal HbA1c	-0.757	2.295	-5.256–3.741	0.742
Time × Group Interactions				
Time × No DM — Prediabetes	0.015	0.149	-0.276–0.307	0.918
Time × No DM — HbA1c unknown	—	—	—	—
Time × New-Onset DM	0.161	0.077	0.010–0.312	0.037 *
Baseline Covariates				
Pre-op HbA1c (per 1%)	1.512	2.443	-3.277–6.300	0.538
Age at surgery (per year)	0.033	0.074	-0.112–0.177	0.657
BMI: Underweight vs Normal	—	—	—	—
BMI: Overweight vs Normal	0.004	2.086	-4.085–4.093	0.999
BMI: Obese vs Normal	2.315	1.997	-1.598–6.229	0.249
Distal pancreatectomy	-0.068	1.849	-3.692–3.557	0.971
Neoadjuvant therapy	-2.137	2.715	-7.458–3.184	0.433
Adenocarcinoma diagnosis	1.443	2.120	-2.712–5.598	0.498
Tobacco use (ever)	-1.137	1.702	-4.472–2.199	0.506

Abbreviations and Notes: β = unstandardized regression coefficient (HbA1c %, change per unit). SE = standard error. CI = 95% confidence interval (Wald). Models fitted by restricted maximum likelihood (REML) using lme4. P-values from Satterthwaite degrees of freedom approximation (lmerTest). * $p < 0.05$; † $p < 0.10$. Time is centered at Year 1 (time_c = 0 corresponds to Year 1 post-op). The β for Time represents the annual HbA1c change in the reference group (No DM — Normal HbA1c). Time × Group interaction terms represent the additional annual change in each group relative to the reference. Model 1: unadjusted, random intercept + slope. Model 2A: adjusted, random intercept, full cohort (pre-op HbA1c excluded due to 56% missingness). Model 2B: adjusted, random intercept, restricted to patients with available pre-op HbA1c. The No DM — Prediabetes group is included in model estimation but suppressed from the trajectory figure due to unreliable estimated marginal means arising from limited post-operative HbA1c observations (n ≈ 50). Reference categories: DM/glycemic group, No DM — Normal HbA1c; BMI, Normal (18.5–24.9 kg/m²). Pre-operative HbA1c ≥6.5% excluded (n = 11, ADA DM criteria).

Table8 FineGray DM

Table 8. Fine-Gray Subdistribution Hazard Model for New-Onset Diabetes Mellitus Following Pancreatectomy (N = 724; Events = 145)

Variable	N	SHR	95% CI	P-Value
BMI: Overweight vs Normal	724	1.70	1.14–2.54	0.010 *
BMI: Obese vs Normal		2.47	1.63–3.74	<0.001 *
BMI: Underweight vs Normal		0.59	0.08–4.54	0.610
Distal pancreatectomy		1.63	1.14–2.34	0.008 *
Adenocarcinoma diagnosis		0.79	0.53–1.18	0.250
Neoadjuvant therapy		0.88	0.54–1.43	0.610
Family history of DM		1.18	0.91–1.53	0.200

Abbreviations and Notes: SHR = subdistribution hazard ratio; CI = confidence interval. Fine-Gray regression accounts for death as a competing event for new-onset diabetes mellitus. * $p < 0.05$; † $p < 0.10$. Reference category: BMI, Normal (18.5-24.9 kg/m²). Patients with pre-operative HbA1c ≥6.5% excluded (n = 11).

Table9 FineGray ExocrineInsufficiency

Table 9. Fine-Gray Subdistribution Hazard Model for Post-Operative Exocrine Insufficiency Following Pancreatectomy (N = 601; Events = 247)

Variable	N	SHR	95% CI	P-Value
Distal pancreatectomy	601	0.50	0.35–0.71	<0.001 *
Adenocarcinoma diagnosis		1.57	1.11–2.22	0.010 *
Neoadjuvant therapy		1.60	1.18–2.16	0.002 *
New-onset diabetes mellitus		1.98	1.50–2.62	<0.001 *

Abbreviations and Notes: SHR = subdistribution hazard ratio; CI = confidence interval. Fine-Gray regression accounts for death as a competing event for exocrine insufficiency. Patients with pre-operative exocrine insufficiency were excluded (n = 123). New-onset diabetes mellitus was included as a time-concurrent covariate reflecting shared pancreatic parenchymal loss. * $p < 0.05$; † $p < 0.10$.